

Dylan Coxshall

Gloucestershire, UK • +44 79560 84982

dcoxshall.github.io | dacoxxshall@gmail.com | linkedin.com/in/dylan-coxshall-3b4695207

BSc Computer Science at the University of Birmingham, with a year in industry as a safety-critical software engineering placement student at GE Aerospace. Certified Engineering Technician by the Institute of Engineering and Technology.

Top Skills

- Safety-critical software engineering in C. Also fluent in Python and Rust, with additional experience in Java, C++, and frontend web development (JavaScript including React.js, HTML and CSS) as well as backend using Flask.
- Embedded software tools and requirements engineering – Lauterbach’s TRACE32, IBM’s Rational DOORS, IBM’s Engineering Workflow Management.
- Experience in Windows and Linux development environments. Experience with the cloud using Hetzner’s cloud hosting service.
- Software development tools – Git and GitHub, and CI/CD using GitLab.

Experience

GE Aerospace, Cheltenham, Gloucestershire, UK (On-site)

Software Engineering Placement Student

June 2024 – July 2025

- Contributed to the secondary power distribution system for a flagship GE Aerospace product, developing, testing, and reviewing safety-critical embedded C code.
- Authored and reviewed software requirements, ensuring compliance with high code quality standards in a safety-critical environment.
- Organised global educational calls with executives and HR leaders as part of the GE Placement Committee, supporting peer development and engagement.
- Led STEM outreach events and became a certified Engineering Technician (See below).

Birmingham Investment and Finance Society, University of Birmingham

Investment Analyst – Technology Sector

October 2025 – Present

- Worked in a team of analysts to guide investments within the technology sector.
- Researched and pitched stocks to an audience, including carrying out analysis on global technology companies.
- Learnt about real-world investment strategies and gained an appreciation for the UK’s financial sector.

Edinburgh Parallel Computing Centre (EPCC), Edinburgh, UK (On-site)

High-Performance Computing Summer School 2025

July 2025

- Learned OpenMPI and the OpenMP API to write programs for EPCC’s Cirrus HPC system.
- Collaborated with fellow software engineers to explore the accessibility of high-performance computing.
- Attended lectures from HPC experts on the future of supercomputing in the UK.

Cleeve School, Bishop's Cleeve (Internship)

Teaching Internship

June 2026

- Interned as a teacher in the computer science/mathematics faculties.
- Planned lessons for high- and low-ability students, as well as for SEND students, and marked maths papers on behalf of other teachers.
- Delivered full lessons covering introductory content across mathematics and computer science.

Personal Projects

dylstatus (Python) – A status-bar generator for the i3 Window Manager.

Dylstatus is a replacement for i3status, and is designed to provide system information to the user on the desktop. Writing dylstatus was my first exposure to interacting with an external API (i3bar Input Protocol) and provided invaluable experience in software engineering. I used the project daily to provide a status bar for my home computing setup, until I moved away from i3wm.

chesslib (C) – A fast chess move generation library and engine, written in C.

chesslib is a C library to generate legal chess moves quickly. It will eventually be a full-fledged chess engine, but is currently in the early stages of development.

chesslib has been used as a testbed for my own minimal unit testing framework – plans are for this unit testing framework to stand alone as a software project.

editor (Rust) – A command-line text editor.

SSHing into a Linux machine (for example) will not typically give you a GUI, so terminal-based editors are particularly important when working in remote environments. To learn Rust, I wrote my own text editor that could be used from the command line. This project introduced me to the principle of trade-offs, and how to consider multiple choices based on future requirements.

Education

BSc Computer Science with Year in Industry, University of Birmingham, UK

Graduated in 2026 (First Class with Honours).

Relevant optional modules taken include Dependable and Distributed Systems, Advanced Networking, and Advanced Cryptography.

Pate's Grammar School, Cheltenham

A-Levels: Mathematics: A*, Physics: A, Computer Science: A, EPQ (Data Privacy): A

Additional Qualifications

Certified Engineering Technician ("EngTech"), Institute of Engineering and Technology (IET)

I earned EngTech while on my industrial placement year at GE Aerospace. Earning the qualification, in my case, meant demonstrating technical competency in controlled low-voltage laboratory environments, as well as demonstrating knowledge of engineering ethics and general good practice.

Duke of Edinburgh (Bronze, Silver and Gold)

Completed all three tiers of the Duke of Edinburgh's Award through Scouts UK, developing leadership, resilience, and teamwork skills.